Dynamic Mutual Learning for Object Detection in Aerial Imagery

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***Abstract*—Object detection in aerial imagery is a pivotal task for various Earth observation systems, composed of two separate, yet interdependent subtasks: classification and localization. However, existing methods face two fundamental limitations: 1) inconsistency in prediction distributions, where these two subtasks lack spatial distribution alignment, and 2) impracticability of cross-scale representations, where fixed-scale representations impede representation capacity and accuracy for objects of varying sizes in aerial scenarios. To overcome these challenges, this article proposes a novel dynamic mutual learning paradigm that synergizes representation-wise and supervision-wise interactions within a unified detection head. It consists of two learning schemes: 1) dynamic learning, which introduces the dynamic routing mechanism to enable cross-scale fine-grained representation aggregation, significantly benefiting representational efficiency and flexibility, and 2) mutual learning, which establishes prediction alignment by explicitly performing subtask-consistent supervision and collaborative optimization. Moreover, within the entire enhanced detection head, these schemes can be jointly optimized and mutually reinforced. Extensive experimental results on different datasets have demonstrated the effectiveness and superiority of this proposed learning paradigm for object detection in aerial imagery, achieving competitive performance in both detection accuracy and computational efficiency.**

***Index Terms*—Dynamic mutual learning, dynamic representations, mutual supervision, object detection, remote sensing (RS).**

I. Introduction

OBJECT detection is a fundamental task in the field of aerial scene interpretation, which has numerous real-world applications, such as wilderness rescue, disaster monitoring, and urban planning [1], [2], [3], [4], [5]. As one of the most representative aerial scenarios, remote sensing (RS) optical images have attracted widespread attention due to their high availability, and the rapid development of RS object detectors based on deep learning architectures, especially convolutional neural networks (CNNs) and Vision Transformers, has transformed the field. These methods range from region-based two-stage approaches [6], [7], [8], [9], [10] to region-free single-stage approaches [11], [12], [13], [14]. Single-stage methods are often preferred in the aerial context due to their desired balance between accuracy and speed.

The primary goal of RS object detection is to simultaneously localize and recognize geospatial objects in satellite and aerial images. This task is naturally formulated as a multitask optimization problem, requiring the joint learning of two subtasks, classification and localization [15], [16], [17]. However, these subtasks are inherently different or even contradictory: classification aims to learn category discrimination or saliency, achieving scale invariance and translation invariance, whereas localization involves predicting bounding boxes that vary according to instance scales and translations. To alleviate these contradictions, existing single-stage RS object detectors [18], [19], [20], [21] typically adopt two separated and fixed detection heads in parallel for classification and localization, respectively. These detection heads process backbone features sequentially at each scale. However, this heuristic design does not fully resolve the aforementioned issues and suffers from the following two critical deficiencies, severely hindering detection accuracy and efficiency.

A. Intrinsic Inconsistency in Prediction Distributions

An ideal RS detector should yield both accurate classification and localization predictions at the same consistent spatial location [22], [23], [24], [25]. This requires well-trained detection heads to guarantee spatial distribution consistency between the two subtasks, that is, the same object region should achieve the highest classification and localization scores simultaneously. For example, previous RS detectors [26], [27], [28] usually assume that an anchor point (for anchor-free detectors) or anchor box (for anchor-based detectors) at the object center will produce the most accurate predictions for both classification and localization. However, this assumption is difficult to fulfill, as the optimal classification location and the most precise localization box frequently occur in different object regions. As illustrated on the right side of Fig. 1, even a well-trained baseline RS detector

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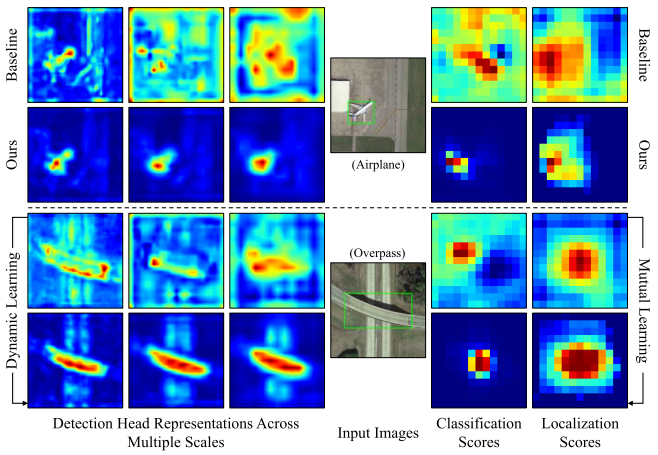


Fig. 1. Illustration of the two critical representational deficiencies in conventional detection heads: subtask prediction inconsistency and cross-scale impracticability, both of which are caused by the design of separated and fixed detection heads. With the proposed dynamic mutual learning, our method yields more accurate and robust head representations and subtask predictions. Dynamic learning enables hierarchical representation aggregation instead of performing predictions using a single scale, while mutual learning aligns the spatial distributions of subtask predictions (classification/localization scores), ensuring that optimal anchor locations simultaneously apply to both subtasks.

typically predicts the maximum classification and localization scores at different positions, such as the upper-left corner and the center of the overpass. These inconsistent predictions may further impede subsequent nonmaximum suppression (NMS), which retains only the bounding box with the highest classification score without considering localization precision. Consequently, predictions with better localization, yet lower classification confidence, are undesirably filtered out, eventually degrading detection accuracy. This inconsistency stems from the separate dual-branch design of the detection heads, which intrinsically disrupts the necessary interaction between the two subtasks. Therefore, mutual knowledge and guidance should be established within the heads to facilitate subtask consistency and alignment. As depicted in the bottom-right part of Fig. 1, the predicted classification and localization score distributions are aligned at identical spatial locations, that is, well-classified anchors should also be well-localized, and vice versa.

B. Cross-Scale Impracticability of Head Representations

Compared to natural-scene objects, RS objects often exhibit large-scale variance due to the bird's-eye capturing mechanism. Consequently, the feature pyramid network (FPN) [29] has played an indispensable role in previous RS detectors, constructing multiscale representations [28], [30], [31], [32], [33]. In conventional detection heads, RS instances of different sizes are separately allocated to the corresponding levels with different feature resolutions. In other words, when perceiving a specific instance, the detection head (including both classification and localization subbranches) only has access to the selected single-scale FPN features. Features from other scales contribute little to the head's prediction. However, encoding every instance based on a single-scale feature leads to adverse or incomplete representation response, suppressing certain discriminative regions. As illustrated on the left side of Fig. 1, for the baseline method, only one scale of representations in the detection head is advantageous and positive, that is, the first scale for the airplane and the third scale for the overpass, while other scales may either be disruptive/confusing or insufficiently activated. This independence and isolation between multiscale representations in the detection head can be considered as cross-scale impracticability, which undermines the learning capacity and flexibility of the head. Moreover, such heads employ fixed and static structures that treat all instances indiscriminately, making it challenging to adapt to the diverse scale distributions in RS scenarios. By contrast, an advanced detection head should enable dynamic cross-scale interaction and modeling to produce more discriminative and robust representations. As shown in the left-bottom part of Fig. 1, in this way, head representations across different scales can be enhanced with better semantics and integrity to jointly benefit subsequent predictions.

To address the deficiencies mentioned above, in this article, we propose a conceptually novel method for object detection in RS images, namely, dynamic mutual learning. As illustrated in Fig. 2, this learning paradigm can be flexibly and seamlessly integrated into the detection head to construct a dynamic mutual space, which facilitates both cross-scale representation-level and spatial prediction-/supervision-level interactions between the two subtasks, that is, classification and localization. Specifically, this space consists of two subspaces for dynamic learning and mutual learning, respectively. In the dynamic subspace, pixel-wise dynamic fusion is established across multiple representation scales. This dynamic mechanism is adaptive to the physical scale and semantic content of the input images. Compared to traditional static detection heads, the dynamic head has the advantage of modeling complex inputs by activating all hierarchical scales instead of being restricted to a single scale, while maintaining high efficiency. In the mutual subspace, the classification-localization prediction distributions are explicitly aligned by establishing a subtask mutual supervision flow, which allows these two subtasks to collaborate more effectively. Simply put, the training samples for classification are defined by localization predictions and supervision rather than relying only on classification supervision, and vice versa. It is worth noting that dynamic learning and mutual learning are not independent but closely interrelated. On the one hand, as illustrated in Fig. 1, the dynamic mutual representations generated by dynamic aggregation can seamlessly bridge the two subspaces, facilitating end-to-end training. On the other hand, dynamic and mutual learning can mutually reinforce each other. For instance, with the dynamic adaptability in pixel-level subtask representations, the establishment of mutual supervision will become more accurate and flexible. Mutual learning, in turn, encourages the selection of more efficient and precise cross-scale pathways in the dynamic subspace, producing more discriminative representations. Comprehensive experiments on different RS object detection datasets demonstrate the effectiveness and superiority of the proposed method. The contributions of this article can be summarized as follows.

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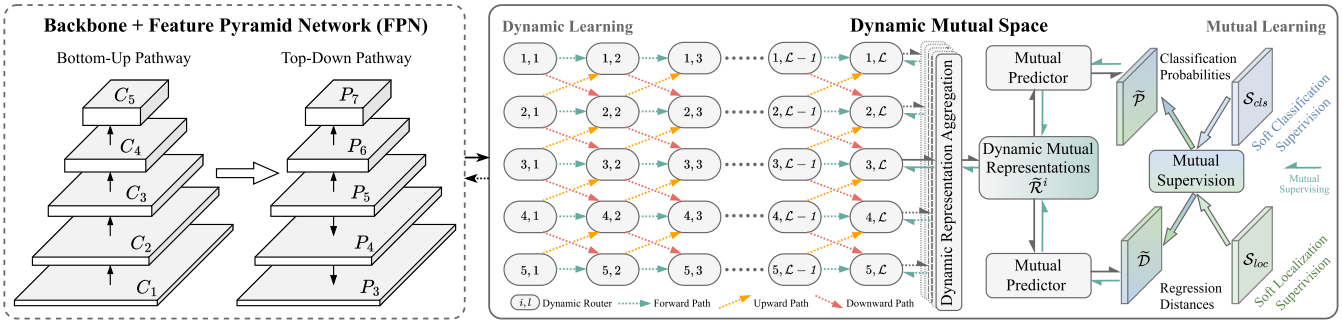


Fig. 2. Overview of the proposed dynamic mutual learning paradigm for RS object detection. The detection framework is composed of a backbone, FPN, and an enhanced detection head to construct a dynamic mutual space. In the backbone and FPN, the upward and downward connection arrows represent downsampling and upsampling operations, respectively. In the dynamic subspace, a routing mechanism facilitates cross-scale path determination and aggregation, with three possible paths for each router: forward, upward, and downward. In the mutual subspace, both soft classification and localization supervision are employed to form mutual supervision, further optimizing dynamic mutual representations for efficient and accurate detection.

1) A novel and advanced learning paradigm, namely, dynamic mutual learning, is systematically devised for RS object detection. This paradigm can flexibly integrate into the detection head to form a dynamic mutual space, enabling cross-scale representation-wise and supervision-wise interactions. This enhanced head addresses the issue of two subtasks being separated and significantly alleviates their prediction conflicts, yielding superior detection accuracy and efficiency.
2) A dynamic learning scheme is proposed to enable the activation and aggregation of fine-grained hierarchical detection representations across multiple scales, addressing the scale variance challenge in RS objects. Moreover, thanks to the proposed dynamic routing mechanism, the dynamic learning-based detection head outperforms fixed architectures in terms of both performance and efficiency.
3) A mutual learning scheme is proposed to align the classification-localization representations, making the subtask prediction distributions more consistent. This strategy tackles the issue of semantic imbalance among different components of RS objects. By computing subtask consistency and mitigating the supervision-level discrepancies, this scheme fosters effective collaboration between the two subtasks and prevents prediction-level conflicts. Additionally, mutual learning features the self-adaptive determination of training samples, eliminating reliance on handcrafted expert knowledge.

The rest of this article is organized as follows. Section II reviews related work, followed by a detailed elaboration of the proposed dynamic mutual learning method in Section III. Extensive experimental results are presented and analyzed in Section IV. Finally, Section V concludes.

II. Related Work

This section briefly reviews three highly relevant topics to this work, that is, RS object detection, training sample assignment strategies, and dynamic networks.

A. RS Object Detection

Inspired by the significant success of deep learning in generic object detection, most modern RS object detectors are based on CNNs and Vision Transformers [11], [34], [35], [36], [37], [38], [39]. According to whether region proposals should be generated in advance, RS object detectors can be divided into two categories: two-stage and single-stage methods. The former, typically motivated by the well-known Faster R-CNN [40], first generates region proposals from CNN features and subsequently performs classification and localization based on these proposals. However, limited by their heavy architectures and inefficiency, two-stage detectors fail to meet the demands of RS scenarios. Instead, single-stage methods [35], [41], [42], [43] have gained more attention for their improved efficiency. For instance, ABNet [35], constructed on the representative single-stage detector, RetinaNet [44], introduced an adaptive balanced network to address the multiscale issues in RS object detection. EIA -Transformer [5] fully leveraged the spatial redundancy of RS images, introducing an efficient Transformer-based framework with reduced computational complexity and higher efficiency. SFS-CNet [45] developed a lightweight convolution module that enhances feature diversity through space-frequency processing to improve the computational efficiency. However, the above single-stage RS detectors, which are primarily anchor-based methods, inevitably rely on tiling numerous anchor-boxes for bounding box regression and category prediction. In contrast, the recent emerging anchor-free detectors [46], [47], [48], [49], [50], [51] directly perform classification and localization on anchor-points within each instance. This eliminates complex handcrafted tuning of anchor boxes and provides compact and simplified detection pipelines with greater flexibility. Anchor-free methods have inspired advancements in RS object detection by potentially addressing extreme object scales and aspect ratios. Recent anchor-free RS detectors [26], [27], [28] further refined this approach. For instance, GLSANet [28], based on the anchor-free YOLOX [50], introduced a global semantic interaction module and a local attention pyramid to address challenges specific to RS scenarios. DiffDet4SAR [52] introduced a novel diffusion model-based framework for target detection, formulating the task as a denoising process of bounding boxes. MaDiNet [53] proposed a Gamma diffusion model network that integrated a Gamma distribution to model image statistics and capture target structures, which yielded improved results in SAR images. Our proposed dynamic mutual learning can seamlessly

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integrate into single-stage RS detectors, both anchor-based and anchor-free methods, enhancing detection accuracy and efficiency.

B. Training Sample Assignment Strategies

Selecting appropriate training samples and assigning them positive or negative labels have been identified to be crucial for detection performance [54], [55], [56]. For anchor-based and anchor-free detectors, training samples are treated as anchor-boxes and anchor-points, respectively. Most anchor-based detectors, including Faster R-CNN [40] and RetinaNet [44], scatter a large number of anchor boxes over the detection head features and then compute their IoUs with ground-truth boxes as the criterion for training sample assignment. In contrast, anchor-free detectors [46], [47], [51], [57] avoid tiling anchor boxes and typically designate anchor points within the center region of an instance as positive samples. Recently, more advanced training sample assignment strategies have been explored to enhance performance. For example, MAL [25] and Noisy-Anchor [58] introduced an anchor-box scoring mechanism based on classification and localization losses. PAA [56] further modeled this procedure as a likelihood maximization for a probability distribution. During inference, anchor boxes independently and directly predict bounding boxes, and those with the highest classification scores are retained after NMS. Unfortunately, without subtask interaction, these methods often suffer from subtask inconsistency or misalignment. Some anchor-based detectors [23] have attempted to introduce mutual information to benefit anchor-box assignment for one subtask by considering the prediction quality on the other subtask. Nevertheless, efficiently incorporating mutual learning into training sample assignment for optimal anchor selection, especially in the RS context, remains an open challenge.

C. Dynamic Networks

Dynamic networks have increasingly drawn research attention for their ability to dynamically adjust network architectures to adapt to different input images or features. Based on dynamic model compression, previous methods focus on dropping blocks [59], [60], [61], [62] and pruning channels [63], [64] to accelerate inference. For instance, SkipNet [60] employed a reinforcement learning-based gating network to skip convolutional blocks. DVT [65] cascaded with an increasing number of Transformer tokens and activated them sequentially for dynamic inference. However, these methods focus only on image classification and are unsuitable for dense prediction tasks, such as object detection and semantic/instance segmentation. Recent solutions [66], [67] have begun bridging this gap. For example, DynaMask [67] introduced a dynamic mask selection network to allocate masks of different resolutions to objects of varying sizes, enhancing segmentation performance. Similarly, DynamicDet [66] improved adaptive feeding [68] by cascading two detectors and a classifier, where the classifier first divides input images into "easy" and "hard" categories and processes them through different detectors. However, this approach remains image-wise dynamic (i.e., all instances within an image are processed by the same and fixed detection architecture) rather than instance-wise dynamic. Given the large-scale variance of RS objects, flexible and robust instance-wise dynamic learning is more favorable for both detection accuracy and efficiency.

III. Dynamic Mutual Learning

This article introduces a novel dynamic mutual learning paradigm for RS object detection, which is characterized by its constructed dynamic mutual space in the enhanced detection head. In this section, we first provide an overview of the framework, followed by interpretations of the dynamic learning and mutual learning schemes. Finally, we introduce the dynamic mutual optimization process with training losses for flexible end-to-end training.

A. Framework Overview

Building on previous single-stage detection methods, such as RetinaNet [44], the proposed dynamic mutual learning paradigm adopts a streamlined "backbone-FPN-head" pipeline, as illustrated in Fig. 2. Such a design aligns with the typical architecture of previous detectors [47], [48]. Specifically, in the FPN, P_6 and P_7 are generated by downsampling P_5 via strided convolutions, thereby enabling the output representations of the FPN to model objects of varying scales. For efficiency and simplicity, it only considers a single anchor at each location, similar to ATSS [48]. Consequently, this design makes the paradigm applicable to both anchor-free and anchor-based detection. In anchor-free detection, the anchor point serves as the anchor, whereas in anchor-based detection, the anchor box plays that role. However, as analyzed earlier, conventional single-stage detection pipelines face two limitations: "cross-scale impracticability" and "subtask prediction inconsistency." These issues stem from the divergence of the two subtasks, classification and localization, which are implemented through two fixed and parallelly separated head branches. To address these two challenges in the RS context, this article innovatively introduces the concepts of "dynamic" and "mutual" learning, which are integrated into the detection head to perform dynamic mutual learning. As depicted in Fig. 2, the enhanced detection head adopts an efficient design to unify classification and localization into a single branch. Dynamic learning and mutual learning collaborate within the dynamic mutual space, enabling cross-scale interaction and distribution alignment of subtask representations, respectively. For flexibility, these two schemes can operate independently or be seamlessly connected through dynamic representation aggregation, enabling end-to-end optimization. Specifically, in the dynamic subspace, fine-grained dynamic routing is performed on the FPN features to produce versatile representations for both subtasks. In the mutual subspace, mutual predictors make coarse predictions based on these representations. These predictions are then refined through soft mutual supervision, with the degree of subtask consistency being signified. During training and backpropagation, classification probabilities and regression (localization) distances are adaptively adjusted based on mutual signals, followed by the

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corresponding updates to the dynamic mutual representations and the routing process.

B. Dynamic Learning

Dynamic learning aims to facilitate feature interaction across different FPN levels, enabling the construction of fine-grained and discriminative representations while significantly improving detection efficiency in the RS context. This is achieved through the proposed dynamic routing mechanism, which balances computational efficiency with performance.

1) Dynamic Routing Space: As shown in Fig. 2, the FPN extracts multilevel features from the backbone and constructs a feature pyramid through a top-down pathway, where low-resolution and high-resolution features are combined. Conventionally, the detection head is positioned after each FPN layer or scale, with parameters shared across all scales, which strives to enhance the semantic and localization capabilities for classification and localization, respectively. However, fixing the detection head to a single FPN scale prevents other-scale representations from contributing directly to subtask predictions. To alleviate this issue, we promote the conventional fixed head to a dynamic head based on dynamic networks [69], [70], which surpasses static networks in both capacity and efficiency. Broadly speaking, the dynamic routing space encompasses all potential paths for routing, while the dynamic routing process aggregates multiscale representations and selects paths based on the input. Specifically, as shown in Fig. 2, the routing space, considered as the dynamic subspace within dynamic mutual space, contains $\mathcal{L}$ depths in the horizontal direction. In the vertical direction, since the FPN typically involves five levels of features, that is, $\{P_3, P_4, P_5, P_6, P_7\}$, the routing space has five distinct scales, that is, $i \in \{1, 2, 3, 4, 5\}$. The core computational unit is a dynamic router that determines the subsequent path, offering up to three candidate paths for each scale transformation, namely, forward, upward, and downward paths, corresponding to keeping, upsampling, and downsampling the scale/resolution, respectively. It is worth noting that this dynamic routing space enhances pixel-level representation abilities, enabling each pixel within the representations to originate from different FPN scales or even utilize multipath, multiscale weighting. This approach eliminates the restriction to a single specific scale, as in conventional settings.

2) Dynamic Routing Process: Within the dynamic subspace (i.e., routing space) containing multiple individual routers, the routing process aims to aggregate multiscale feature representations and choose pixel-level routing paths.

a) Dynamic routers and dynamic gates: The dynamic routing mechanism at the router level is illustrated in Fig. 3(a). Given the element-wise accumulation of three features with different spatial sizes from the former layer $l-1$, denoted as $\mathbf{Y}^{i-1,l-1}$, $\mathbf{Y}^{i,l-1}$, and $\mathbf{Y}^{i+1,l-1}$, respectively, the input of the lth layer is formulated as $\mathbf{X}^l \equiv \{\mathbf{x}_k^l\}_{k=1}^{\mathcal{K}} = \mathbf{Y}^{i-1,l-1} + \mathbf{Y}^{i,l-1} + \mathbf{Y}^{i+1,l-1}$, where $\mathcal{K}$ represents the total number of pixel-level locations. Given the alternative paths $\mathbf{P}^{i,l} = \{p^{i,l,n}(\cdot) | n \in \{1, \ldots, \mathcal{N}\}\}$, inside each router, $\mathcal{N}$ different dynamic gates are devised to dynamically control the subsequent operations, for example, ON–OFF states of a path, for each pixel-level location. The

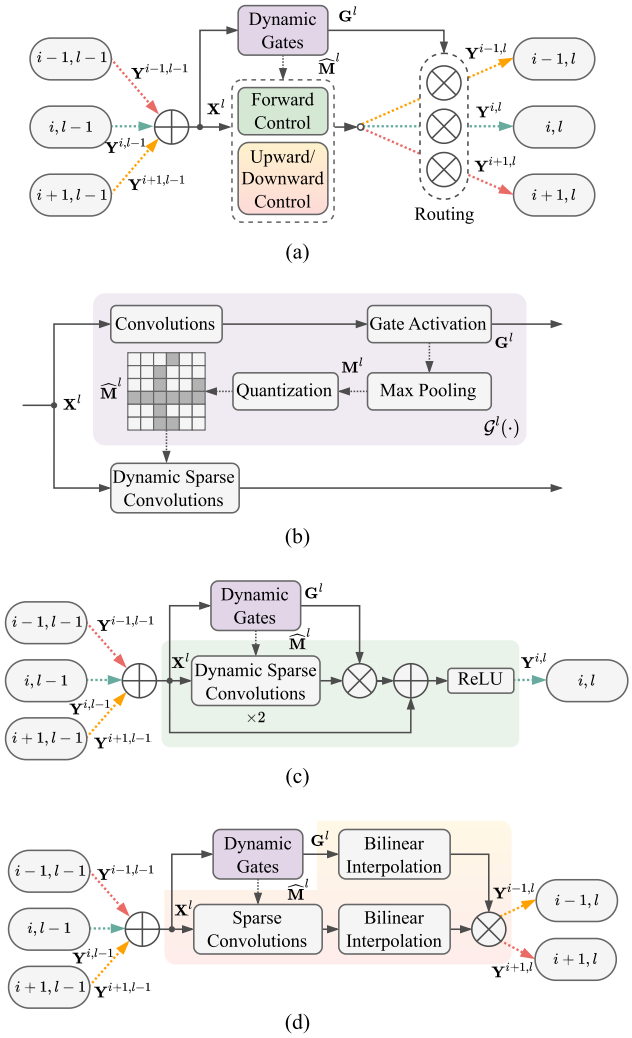


Fig. 3. Illustration of the components in the proposed dynamic learning scheme. (a) Dynamic routing mechanism within a dynamic router, involving three candidate paths. (b) Schematic of the dynamic gate, which generates a pixel-wise mask for dynamic sparse convolutions. (c) Structure of the dynamic forward path for maintaining scales. (d) Structure of the dynamic upward/downward paths for scaling up/down.

gate output, which serves as the activating factor, is pixel-wise with respect to k, calculated as follows (omitting i for simplicity):

$$g_k^{l,n} = \mathcal{G}^{l,n}\left(\mathbf{x}_k^l; \mu^{l,n}\right) \in [0, 1] \tag{1}$$

where $\mathcal{G}^{l,n}(\cdot)$ represents the nth dynamic gating function and $\mu^{l,n}$ refers to its corresponding parameters shared across all locations. To enable flexible multipath selection and aggregation in dynamic routing, we follow [70], [71] and relax the activating factors $\mathbf{G}^l \equiv \{g^{l,n}\}_{n=1}^{\mathcal{N}}$ into a soft continuous space, that is, $[0, 1]$, instead of the hard discrete space $\{0, 1\}$ in previous works [60], [72]. This relaxation facilitates balancing diverse paths via multiple weights to determine the routing output, which is formulated as follows:

$$\mathbf{y}_k^l = \left\{\mathbf{y}_k^{l,n} | \mathbf{y}_k^{l,n} = g_k^{l,n} \cdot p^{l,n}\left(\mathbf{x}_k^l\right), n \in \mathbf{N}_k\right\} \tag{2}$$

where $\mathbf{N}_k = \{n | g_k^{l,n} > 0, n \in \{1, \ldots, \mathcal{N}\}\}$, representing that these candidate paths with positive activating factors

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will be activated simultaneously in the router. As illustrated in Fig. 3(b), such dynamic gates $\mathcal{G}^l(\cdot)$ are instantiated by lightweight convolutions, whose input is compressed into a single-channel representation, followed by the proposed gate activation function to implement pixel-wise path control.

b) Dynamic gate activation: The routing likelihood of each subsequent path, that is, its probability of being selected, is handled by the dynamic gating function $\mathcal{G}^l(\cdot)$, while its internal gate activation function directly determines the quality of path selection and computational efficiency during training. Typically, both hard nondifferentiable functions (e.g., Sigmoid [73]) and soft differentiable functions (e.g., Softmax [71] and ReTanh [58]) can be employed for gate activation. However, these methods usually require complicated additional thresholding during inference or lack excellent continuous monotonicity within the range $[0, 1]$, leading to significant gradient variations during training. Therefore, to further boost the learning ability and stability of the dynamic routing process, we develop an advanced differentiable gate activation function based on the Tanh$(\cdot)$ function as follows:

$$\Gamma(j) = \max\left(0, \frac{\text{Tanh}(j-\delta) + \text{Tanh}(\delta)}{1 + \text{Tanh}(\delta)}\right) \in [0,1] \quad \forall j \in \mathbb{R} \tag{3}$$

where δ represents the hyperparameter regulating the gradient. Assigning a positive value to δ mitigates the discontinuity issue by decreasing the gradient at zero.

c) Dynamic paths for scale transformations: Since the activating factor $g^{l,n}$ is affected by the input $\mathbf{x}^l$ in (1), the dynamic routing process is considered data-dependent. As depicted in Fig. 3(a), three dynamic routing paths can be selected for each router, namely, the "forward path," "upward path," and "downward path," whose internal operations are illustrated in Fig. 3(c) and (d). For the forward path in Fig. 3(c), which aims to keep the representation scale unchanged, dynamic sparse convolutions [74] are adopted to enhance routing efficiency. In Fig. 3(d), both upward and downward paths include the bilinear interpolation for scaling up or down, that is, upsampling or downsampling, with the scale factor set to 2, and depthwise convolutions [75] are leveraged for efficiency enhancement. The previously defined $\mathbf{P}^{\hat{i},l}$ actually corresponds to these dynamic scale transformations, where $\hat{i} \in \{i-1, i, i+1\}$ implies these three options for scale transformations. Additionally, as shown in Fig. 3(b), a fine-grained pixel-wise mask $\widehat{\mathbf{M}}^l$ is produced based on $\mathbf{G}^l$ through max pooling followed by quantization, which can determine the exact discrete locations where sparse convolutions need to be executed, significantly reducing the computational load. It is worth noting that the intermediate masks $\mathbf{M}^l \equiv \{m^{l,n}\}_{n=1}^{\mathcal{N}}$ will be utilized in computational efficiency expectation below.

3) Computational Efficiency Expectation: One desirable benefit of our dynamic learning scheme is the excellent trade-off between accuracy and efficiency, which alleviates the issue of limited computational resources in practical RS systems. Each dynamic router should be encouraged to deactivate those paths with high complexity but negligible performance loss. To achieve this, the computational efficiency expectation strategy is proposed to budget the computational burden for a more efficient routing process. Given the computational complexity $\Omega^{l,n}$ for the routing path n of the router l, the computational efficiency expectation is calculated as follows:

$$\mathcal{E}^l = \sum_{k}^{\mathcal{K}} \sum_{n}^{\mathcal{N}} \Omega^{l,n} m_k^{l,n}, \text{where } m_k^{l,n} = \max_{k' \in \mathbf{N}_k^{l,n}} \left(g_{k'}^{l,n}\right) \tag{4}$$

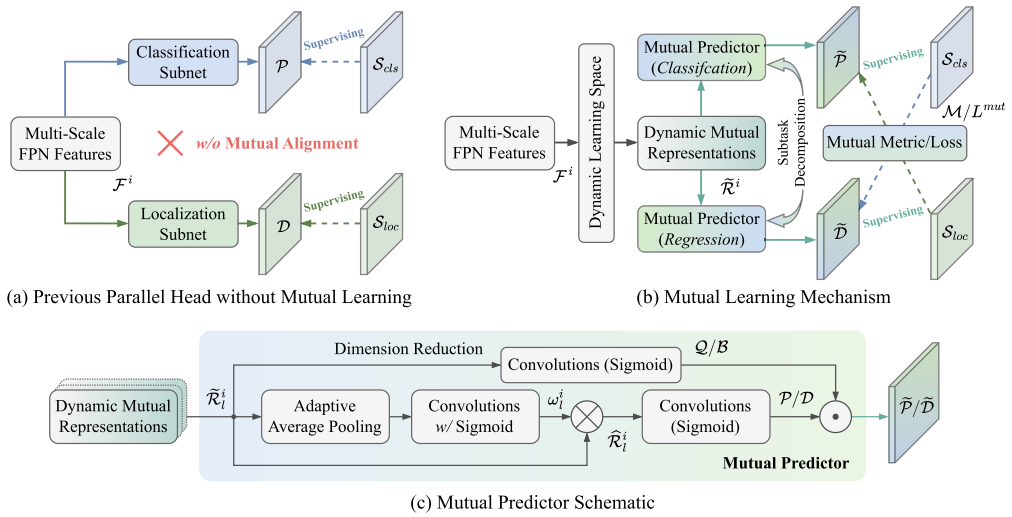
and $\mathbf{N}_k^{l,n}$ typically refers to the receptive field at the kth location in the nth path, which tends to increase with depth. Notably, only the locations with positive activating factors are enabled in the mask $m^{l,n}$ for calculations, where the max operation further promotes complete and precise connected regions. This efficiency expectation can dramatically reduce computational complexity and latency by being seamlessly integrated into the end-to-end dynamic mutual optimization (see Section III-D).

C. Mutual Learning

At the end of the dynamic subspace, hierarchical representations from different levels or scales are aggregated along the channel dimension. This aggregation enhances discriminability and bridges the two subspaces. As illustrated in Fig. 2, the mutual subspace follows the dynamic subspace, together constituting the dynamic mutual space. While dynamic learning aims to efficiently develop cross-scale representations applicable to both subtasks, that is, focusing on the representation level, mutual learning is intended to explicitly align the prediction distributions of the two subtasks, directly executing at the prediction and supervision level. This goal is achieved through two aspects: structurally strengthening the subtask interaction (see Section III-C1) and enhancing the subtask alignment via the definition of supervisory signals (see Section III-C2). Additionally, since mutual learning primarily takes effect during training to foster subtask consistency, the learned detector can achieve performance gains without incurring additional computational costs during inference.

1) Mutual Prediction: As illustrated in Fig. 4(a) and (b), the proposed mutual learning mechanism replaces the dual-branch design, differing structurally from the traditional parallel detection heads without mutual alignment. It can be observed that both types rely on multiscale FPN features, while mutual learning-related components are positioned after dynamic learning, operating directly on the dynamic mutual representations $\widetilde{\mathcal{R}}^i$. In other words, during the construction of $\widetilde{\mathcal{R}}^i$, subtask interaction has already commenced, indicating that dynamic learning and mutual learning are not independent, yet the representation–interaction process is still driven primarily by mutual learning. Structurally, inspired by [76], [77], dynamic mutual representations followed by two mutual predictors in Fig. 4(b) jointly serve the roles of subtask subnets in Fig. 4(a). However, if both subtask predictions, classification and localization, are performed directly on the representation $\widetilde{\mathcal{R}}^i$, subtask conflicts will be inevitably introduced due to the single-branch design [76]. Consequently, the mutual predictors can conveniently provide a buffer to mitigate representation conflicts and efficiently connect mutual supervision with $\widetilde{\mathcal{R}}^i$ by conducting subtask decomposition. Specifically, the proposed mutual predictor schematic is interpreted in Fig. 4(c), where these two mutual predictors for classification and regression

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(a) Previous Parallel Head without Mutual Learning

(b) Mutual Learning Mechanism

(c) Mutual Predictor Schematic

Fig. 4. Illustration of the proposed mutual learning scheme. (a) Comparison with a traditional parallel detection head design without mutual learning. (b) Mutual learning mechanism that establishes subtask decomposition with mutual metrics. (c) Schematic of the mutual predictor, which features a highly lightweight and transferable structure.

share the same architecture. Given the dynamic mutual representation $\widetilde{\mathcal{R}}_l^i$ at the lth layer, ω_l^i can be generated through adaptive average pooling and convolutions, which aim to aggregate cross-layer dependencies to further produce subtask representations

$$\widehat{\mathcal{R}}_l^i = \omega_l^i \widetilde{\mathcal{R}}_l^i \quad \forall l \in \{1, 2, \ldots, \mathcal{L}\}. \tag{5}$$

Then, the initial classification prediction $\mathcal{P}$ and regression prediction $\mathcal{D}$ are generated through convolutions based on $\widehat{\mathcal{R}}^i$. These predictions are further explicitly aligned by addressing their distribution inconsistency. Specifically, the spatial probability map $\mathcal{Q}$ and offset map $\mathcal{B}$ are computed based on the dynamic mutual representations $\widetilde{\mathcal{R}}^i$ using dimension reduction and convolutions. These maps are utilized for pixel-wise subtask alignment, as follows:

$$\widetilde{\mathcal{P}} = \mathcal{P} \odot \mathcal{Q} = \sqrt{\mathcal{P} \times \mathcal{Q}} \tag{6}$$

$$\widetilde{\mathcal{D}} = \mathcal{D} \odot \mathcal{B} = \mathcal{D}(k + \mathcal{B}(k)) \tag{7}$$

where $\odot$ represents element-wise adjustment for classification or localization. The localization adjustment in (7) for each location k is implemented through linear interpolation. It is worth noting that the mutual predictor for subtask decomposition is highly lightweight, incomparable to the scale of the subtask subnets in Fig. 4(a). Its learning procedure is guided by the mutual supervision, which is detailed in the following.

2) Mutual Supervision: A high-quality anchor should be optimal for the predictions of both subtasks, that is, it should have high classification probabilities and low regression distances simultaneously. Anchors with high scores for only one subtask should be given relatively lower priority during training. To this end, we establish a bag of anchor candidates by filtering out background anchors, based on which mutual supervision is performed. Given the classification score $\mathcal{S}_{\text{cls}}$ and localization (IoU-related) score $\mathcal{S}_{\text{loc}}$, a natural approach to construct mutual supervision is toadopt $\mathcal{S}_{\text{cls}}$ to guide supervision for the localization prediction $\widetilde{\mathcal{D}}$, and $\mathcal{S}_{\text{loc}}$ to supervise the classification prediction $\widetilde{\mathcal{P}}$. However, such straightforward mutual supervision may lead to training instability and confuse detectors, especially during the initial stage. As depicted in Fig. 4(b), a joint-likelihood mutual metric is calculated to evaluate how well an anchor serves as a training sample for the ground-truth instance, considering both subtasks simultaneously. The basic form of the mutual metric is $\mathcal{S}_{\text{cls}} \cdot \mathcal{S}_{\text{loc}}$. After applying high-order regularization criteria, it can be formulated as follows:

$$\mathcal{M} = \mathcal{S}_{\text{cls}}^{\mu} \cdot \mathcal{S}_{\text{loc}}^{\nu} \tag{8}$$

where $\mathcal{M}$ indicates anchor-level consistency for each instance and μ and ν are hyperparameters that regularize the subtask impacts in the mutual scheme. As mentioned above, the mutual supervision defined by $\mathcal{M}$ here plays an important role in the joint optimization and learning of the predictions $\widetilde{\mathcal{P}}$ and $\widetilde{\mathcal{D}}$, aiming for subtask alignment. During training sample assignment, anchors with higher rankings (i.e., with larger $\mathcal{M}$) are treated as positive samples for both classification and localization, prioritizing well-aligned anchors while reducing the impact of subtask-inconsistent anchors. Additionally, the mutual metric is integrated into the loss functions described in Section III-D.

D. Dynamic Mutual Optimization

The entire detection framework with dynamic mutual space, illustrated in Fig. 2, can be trained conveniently and flexibly in an end-to-end manner using the following proposed dynamic mutual optimization strategy. The total loss function is formulated as

$$L = L^{\text{dyn}} + L^{\text{mut}} \tag{9}$$

where L^{dyn} and L^{mut} correspond to the optimization losses for dynamic learning and mutual learning, respectively. Specifically, L^{mut} involves the loss functions of the two

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subtasks, that is, classification and localization, computed as follows:

$$L^{\text{mut}} = \lambda_{\text{cls}} L_{\text{cls}}\left(\widetilde{\mathcal{P}}, \mathcal{M}\right) + \lambda_{\text{loc}} L_{\text{loc}}\left(\widetilde{\mathcal{D}}, \mathcal{M}\right) \tag{10}$$

where L_{cls} is the classification loss, which can be implemented using focal loss [44] and L_{loc} is the localization loss, which can be implemented using GIoU loss [78] by default. The coefficients λ_{cls} and λ_{loc} weight the impact of the subtasks. It is worth noting that L_{cls} and L_{loc} are not limited to specific types of loss functions; they can be arbitrary losses. In other words, this mutual learning scheme includes a hyper-loss formulation to enhance subtask interaction and mutual optimization. This allows mutual supervision to contribute to the selection and generation of high-quality bounding boxes for both classification and localization, instead of introducing new losses. As discussed in Section III-B3, different from L^{mut}, L^{dyn} is not directly related to subtask predictions, which aims to incorporate computational efficiency expectation to strike an accuracy–efficiency tradeoff. Specifically, the expectation computed using (4) is adopted to constrain the overall computational cost within the dynamic subspace

$$L^{\text{dyn}} = \gamma \cdot \frac{\sum_l^{\mathcal{L}} \mathcal{E}^l}{\sum_l^{\mathcal{L}} \Omega^l}, \text{ where } \Omega^l = \sum_n^{\mathcal{N}} \Omega^{l,n} \tag{11}$$

where γ is a positive hyperparameter that controls the expectation scale and balances the two learning schemes during training. Notably, without L^{dyn}, dynamic learning can still function, yet may lead to an increased number of activated dynamic paths, focusing only on accuracy improvements. Combining L^{dyn} and L^{mut} in (9) enables the joint optimization of dynamic learning and mutual learning within a unified end-to-end framework.

IV. Experiments and Analysis

Comprehensive experiments were conducted on different RS object detection datasets to evaluate the proposed method. The datasets, evaluation metrics, and implementation details are briefly introduced first, followed by a detailed presentation and analysis of all experimental results, including ablation studies.

A. Datasets

To evaluate the proposed dynamic mutual learning paradigm, two RS object detection datasets, namely, DIOR [17] and NWPU VHR-10 [6], are utilized in the experiments.

1) DIOR Dataset: DIOR [17] is a large-scale and extensively adopted dataset for object detection in RS scenarios. It contains 23 463 images with a total of 190 288 instances across the following 20 geospatial categories: airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DAM), expressway toll station (ETS), expressway service area (ESA), golf field (GF), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill (WM). In the experiments, 1/3, 1/6, and 1/2 of the original images are randomly selected for training, validation, and testing, respectively.

2) NWPU VHR-10 Dataset: NWPU VHR-10 [6], [79] is another popular RS object detection dataset, consisting of 800 very-high-resolution (VHR) RS images, with 650 positive samples and the remaining negatives. This dataset features ten different geospatial categories: airplanes (AP), ships (SH), storage tanks (ST), baseball diamonds (BD), TCs, BCs, GTFs, harbors (HA), bridges (BR), and vehicles (VE). In the experiments, 3/4 of the positive images are randomly selected for training, with the remainder reserved for testing.

B. Evaluation Metrics and Implementation Details

To quantitatively evaluate the performance of different methods, average precision (AP) is employed as the basic evaluation metric, based on calculating the precision rate (P) and recall rate (R). Mean AP (mAP), which represents the mean of APs across all categories, is used for multiclass object detection, calculated as $\text{mAP} = (1/\mathcal{C}) \sum_{c=1}^{\mathcal{C}} \text{AP}_c$, where $\mathcal{C}$ represents the number of object categories. Higher mAP values generally indicate better detection performance. Furthermore, mAPs with different IoU thresholds, that is, mAP_{50}, mAP_{75}, and $\text{mAP}_{50:95}$, are reported. Given that the proposed dynamic mutual learning paradigm aims to boost both detection efficiency and accuracy, we report the number of parameters (abbreviated as Params) and floating-point operations (FLOPs) to evaluate and compare the computational efficiency of various detectors.

As described earlier, the detection pipeline follows a "backbone-FPN-head" structure, utilizing backbones of different sizes, from CNNs to Vision Transformers. Unless specified otherwise, ResNet50 [80] is adopted as the default backbone, with initial weights pretrained on ImageNet [81]. The experiments were mainly conducted on the anchor-free ATSS [48] framework, which assigns only one anchor per location. The simplicity of ATSS is advantageous in highlighting the properties of our dynamic mutual learning paradigm. All the models were trained on the NVIDIA Tesla V100 GPUs with a batch size of 4 for 12 epochs. The AdamW [82] optimizer was used during training, with an initial learning rate of 1×10^{-3} and the weight decay of 1×10^{-3}, and a warm-up strategy is also implemented.

C. Ablation Studies

We conducted comprehensive ablation studies on the DIOR validation set to verify the effectiveness of the proposed dynamic mutual learning paradigm for RS object detection.

1) Dynamic/Mutual Subspace Analysis: The constructed dynamic mutual space integrates two subspaces, that is, the dynamic subspace and the mutual subspace. Each contributes uniquely to the overall detection performance and can be implemented individually. To evaluate their effectiveness and investigate their roles, these subspaces are added separately to the baseline model. As shown in Table I, both dynamic and mutual models achieve superior detection performance compared to the baseline in most cases. Particularly, the dynamic model not only improves accuracy but also significantly enhances inference efficiency, reducing about 117.6G FLOPs, which constitutes a substantial portion of the original computational load. This efficiency gain is attributed to the proposed

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TABLE I

PERFORMANCE EVALUATION AND COMPARISON OF THE BASELINE MODEL, THE PROPOSED DYNAMIC MUTUAL LEARNING, AND ITS TWO SUBCOMPONENTS: DYNAMIC LEARNING AND MUTUAL LEARNING

Methods	APL GTF	APO HA	BF OP	BC SH	BR STA	CH STO	DAM TC	ESA TS	ETS VE	GF WM	mAP_{50} (ΔmAP_{50})	mAP_{75} (ΔmAP_{75})	$mAP_{50:95}$ ($\Delta mAP_{50:95}$)	Params (M)	FLOPs (G)
Baseline	91.1 72.0	63.4 30.4	92.2 45.3	77.8 73.4	22.4 85.0	86.3 79.8	40.8 86.6	56.4 26.8	47.9 65.4	69.4 69.2	64.1 (—)	43.4 (—)	40.9 (—)	36.7	189.6
Dynamic Learning	91.4 73.5	69.8 35.2	92.9 56.0	81.3 **78.8**	34.2 87.0	90.1 81.1	49.0 **88.6**	63.2 48.6	52.0 71.1	74.2 **76.8**	69.7 (+5.6)	49.9 (+6.5)	46.5 (+5.6)	**27.8**	**72.0**
Mutual Learning	90.8 74.0	71.1 42.3	91.6 53.0	**82.5** 76.4	29.4 81.4	90.6 79.8	48.7 88.3	62.9 43.8	54.8 69.0	74.7 67.1	68.6 (+4.5)	49.2 (+5.8)	45.6 (+4.7)	33.4	130.1
Dynamic Mutual Learning	**92.3** **77.6**	**79.1** **45.5**	**93.9** **62.0**	82.4 78.2	**39.8** **87.9**	**91.2** **81.8**	**56.2** 88.5	**65.8** **53.4**	**57.4** **73.6**	**76.1** 76.7	**73.0** (**+8.9**)	**56.0** (**+12.6**)	**51.7** (**+10.8**)	28.8	78.0

TABLE II

PERFORMANCE COMPARISON OF DYNAMIC LEARNING IN DIFFERENT CONFIGURATIONS WITH FIXED LEARNING, WHERE "FLOPS AVG.," "FLOPS MAX.," AND "FLOPS MIN." REPRESENT THE AVERAGED, MAXIMUM, AND MINIMUM FLOPS OF THE MODEL, RESPECTIVELY

Methods		Dynamic	Fine-Grained	mAP_{50}	mAP_{75}	$mAP_{50:95}$	Params (M)	FLOPs Avg. (G)	FLOPs Max. (G)	FLOPs Min. (G)
Fixed Learning	$\mathcal{L}=2$	×	×	61.2	36.7	35.7	**27.3**	**66.3**	**66.3**	**66.3**
	$\mathcal{L}=4$	×	×	62.4	35.6	35.1	27.4	69.9	69.9	69.9
	$\mathcal{L}=8$	×	×	**64.6**	**42.0**	**40.7**	27.7	77.1	77.1	77.1
Dynamic Learning	$\mathcal{L}=2$	✓	×	62.7	40.3	38.3	**27.3**	69.4	**75.5**	67.5
		✓	✓	64.9 (+2.2)	42.6 (+2.3)	40.2 (+1.9)	27.3	**67.0**	77.6	**64.9**
	$\mathcal{L}=4$	✓	×	66.0	44.0	41.2	27.5	72.5	77.4	70.4
		✓	✓	67.6 (+1.6)	46.5 (+2.5)	43.2 (+2.0)	27.5	71.4	82.0	66.2
	$\mathcal{L}=8$	✓	×	69.0	48.6	45.4	27.8	79.3	80.4	68.7
		✓	✓	**69.7** (+0.7)	**49.9** (+1.3)	**46.5** (+1.1)	27.8	72.0	86.3	67.4

dynamic routing mechanism, which optimizes computational resources by adaptively activating paths based on input characteristics. Meanwhile, the mutual model boosts detection accuracy, bringing a notable + 4.5% mAP_{50} gain while reducing the model size of 33.4M parameters and 130.1G FLOPs compared to the baseline. This demonstrates its effectiveness in aligning subtask predictions without increasing complexity. The last row of Table I presents the performance of the full dynamic mutual model, which substantially outperforms the baseline across all aspects, including different accuracy metrics and computational efficiency. It achieves an mAP of 73.0%, representing an 8.9% improvement over the baseline, while maintaining a compact model size of 28.8M parameters and requiring only 78.0G FLOPs for computation. It is worth noting that there is no linear correlation between the FLOPs of these models, as the proposed dynamic learning adapts dynamically, with computational complexity varying depending on input characteristics and model structures. Additionally, we plot the precision–recall (PR) curves in Fig. 5 for these four models, where in most cases, the proposed dynamic mutual learning-based framework consistently achieves superior performance. Based on the above observations, the effectiveness of both dynamic learning and mutual learning schemes has been verified. Integrating both dynamic learning and mutual learning remarkably enhances both accuracy and efficiency for RS object detection.

a) Visualization of aggregated representations: Exemplar visualizations of aggregated representations produced by the four models, that is, the baseline, dynamic learning only, mutual learning only, and dynamic mutual learning, are compared in Fig. 6. The baseline model generates relatively confusing representations, which partially highlight regions of interest but lack precision. Using the proposed dynamic mutual learning improves precision, demonstrating more focused patterns, for example, the circular patterns in the top row and the ship in the bottom row are more clearly defined. The mutual learning scheme provides distinct yet enhanced representation distributions. By combining their strengths, the dynamic mutual learning model yields the most accurate, complete, and well-defined representations, capturing both shapes and structures with high clarity. This performance significantly surpasses that of both the baseline and the individual learning schemes.

2) Effect of Dynamic Fine-Grained Routing: Table II compares the performance of dynamic learning against fixed learning, in terms of both detection accuracy and computational efficiency. Overall, dynamic learning demonstrates superior performance compared to fixed learning, with negligible increases in computational cost or even reductions. For fixed learning, increasing the subspace depth $\mathcal{L}$ from 2 to 8 steadily improves detection accuracy, for example, mAP_{50} rises from 61.2% to 64.6%. A similar trend is observed for dynamic learning, validating that a deeper subspace has better learning capacity. However, these improvements are associated with consistently higher computational costs. Notably, the key advantage of dynamic learning lies in its adaptability and flexibility in computational resource utilization. While fixed learning maintains constant FLOPs regardless of input

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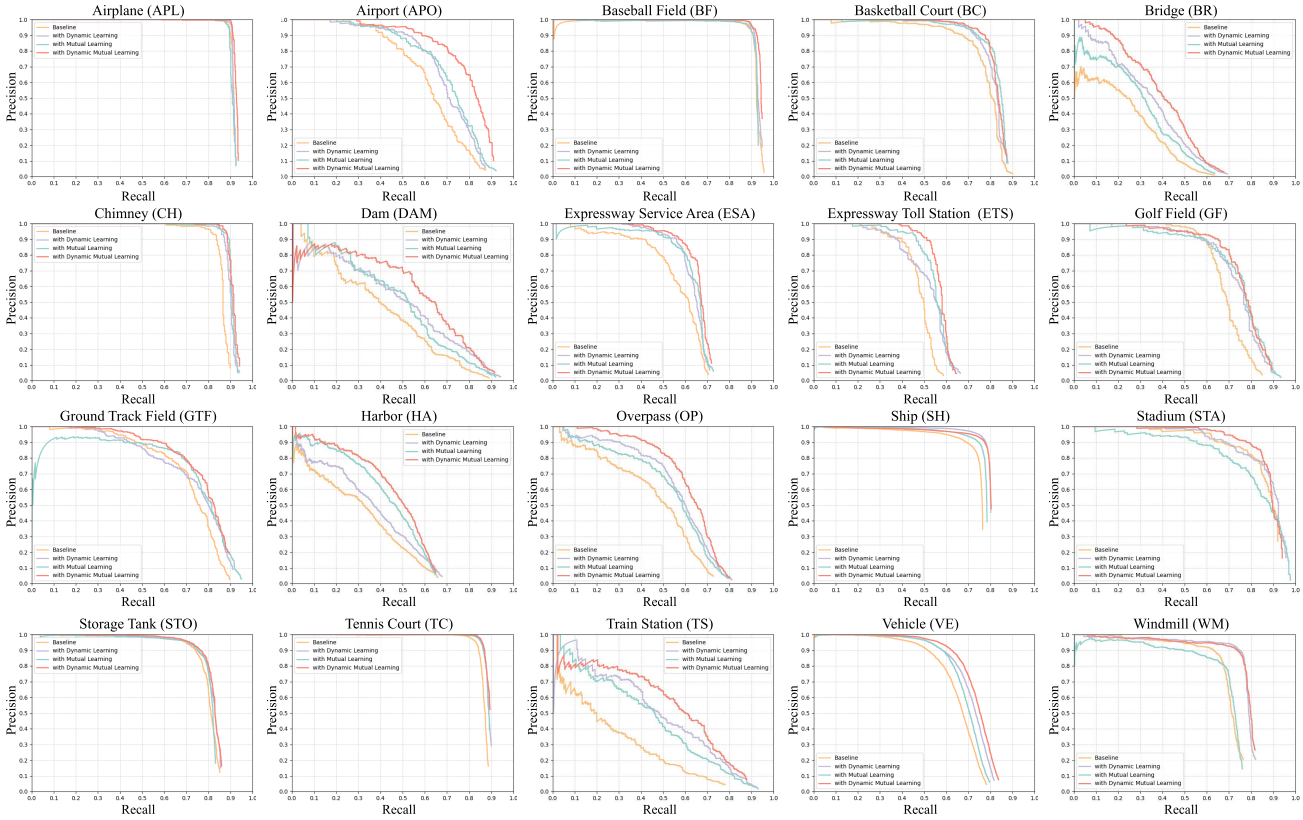


Fig. 5. PR curves of different categories generated by four different frameworks, that is, the baseline, dynamic learning only, mutual learning only, and dynamic mutual learning.

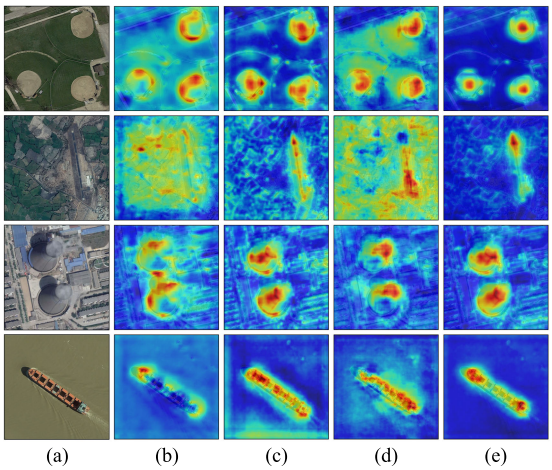


Fig. 6. Visualization of aggregated representations from four different detection frameworks. (a) Input images. (b) Baseline model. (c) Dynamic learning only. (d) Mutual learning only. (e) Dynamic mutual learning.

complexity, dynamic learning dynamically adjusts routing paths according to the input characteristics, accounting for diverse RS scene contents. In addition, the effectiveness of fine-grained pixel-level operations in dynamic routing is verified by replacing this characteristic with global average pooling. Without fine-grained routing, although dynamic models still outperform fixed models, the accuracy gains decrease while computational complexity increases. When $\mathcal{L} = 8$, dynamic learning achieves optimal detection accuracy across all mAP metrics, with lower average computational costs compared to fixed learning (72.0G versus 77.1G FLOPs). This configuration is adopted as the default setting in the experiments. These findings underscore the superiority of dynamic fine-grained routing in optimizing both accuracy and efficiency for RS object detection.

TABLE III
COMPARISON OF DIFFERENT GATE ACTIVATION FUNCTIONS

Activation	δ	mAP_{50}	mAP_{75}	$mAP_{50:95}$	FLOPs Avg.	FLOPs Max.	FLOPs Min.
Sigmoid	–	65.5	44.2	41.5	115.9G	144.7G	94.3G
ReTanh	–	68.8	46.4	44.5	70.3G	77.8G	66.2G
Dynamic Gate Activation (Ours)	0.5	67.6	45.1	43.3	72.3G	86.8G	66.8G
	1.0	69.0	50.1	46.4	73.7G	91.5G	67.7G
	1.5	**69.7**	49.9	46.5	72.0G	86.3G	67.4G
	2.0	69.5	**51.0**	**47.1**	73.1G	91.4G	67.5G

3) Effect of Dynamic Gate Activation: Table III presents a comparison of the proposed dynamic gate activation for dynamic gating against two commonly used activation functions. Specifically, the Sigmoid and ReTanh activations achieve $mAPs_{50}$ of 65.5% and 68.8%, respectively, which are inferior to the performance of the proposed dynamic gate activation. Across various hyperparameter settings, the optimal results are achieved when $\delta = 1.5$ (producing the highest mAP_{50} and serving as the default setting) and $\delta = 2.0$ (producing the highest mAP_{75} and $mAP_{50:95}$). Notably, the values of δ do not significantly impact computational complexity, demonstrating

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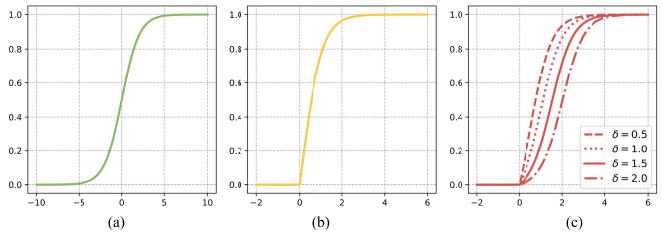


Fig. 7. Visualization of three different activation functions, and the shape of our proposed dynamic gate activation is determined by the hyperparameter δ. (a) Sigmoid. (b) Restricted Tanh. (c) Ours.

TABLE IV

COMPARISON OF DYNAMIC PATHS FOR SCALE TRANSFORMATIONS, WHERE "FW" AND "UD" REFER TO THE DYNAMIC FORWARD PATH AND UPWARD/DOWNWARD PATH, RESPECTIVELY

Methods	FW	UD	mAP_{50}	mAP_{75}	$mAP_{50:95}$	FLOPs Avg.	FLOPs Max.	FLOPs Min.
Baseline	–	–	64.1	43.4	40.9	189.6G	189.6G	189.6G
Dynamic Learning	×	×	64.6	42.0	40.7	77.1G	77.1G	77.1G
	×	✓	67.8	44.5	42.7	87.8G	93.5G	81.3G
	✓	×	68.2	48.9	45.0	74.0G	100.3G	66.8G
	✓	✓	**69.7**	**49.9**	**46.5**	**72.0G**	**86.3G**	**67.4G**

TABLE V

PERFORMANCE OF DYNAMIC LEARNING FOR DIFFERENT VALUES OF THE HYPERPARAMETER γ

Methods	γ	mAP_{50}	mAP_{75}	$mAP_{50:95}$	FLOPs Avg.	FLOPs Max.	FLOPs Min.
Baseline	–	64.1	43.4	40.9	189.6G	189.6G	189.6G
Dynamic Learning	0	69.0	47.9	45.3	87.8G	111.8G	72.5G
	0.1	**69.7**	**49.9**	**46.5**	72.0G	86.3G	67.4G
	0.5	68.9	47.4	44.6	65.0G	71.1G	**63.9G**
	0.8	68.7	47.9	44.8	**64.7G**	**65.1G**	64.4G

TABLE VI

ABLATION STUDY OF MUTUAL PREDICTION AND MUTUAL SUPERVISION IN MUTUAL LEARNING, WHERE "DL" DENOTES THE PROPOSED DYNAMIC LEARNING

Methods	mAP_{50}	mAP_{75}	$mAP_{50:95}$	FLOPs
Baseline	64.1	43.4	40.9	189.6G
Only Mutual Supervision	65.7	44.7	41.9	135.2G
Complete Mutual Learning	68.6	49.2	45.6	130.1G
DL + Mutual Supervision	71.0	51.0	47.6	93.4G
Dynamic Mutual Learning	**73.0**	**56.0**	**51.7**	**78.0G**

the desired stability in enhancing detection efficiency. In Fig. 7, the diagrams of three activation functions are plotted for visual comparison. It is noticeable that the proposed dynamic gate activation varies with δ, becoming progressively smoother as δ increases, suggesting enhanced steadiness and flexibility for dynamic routing. Overall, these observations validate the effectiveness and superiority of our dynamic gate activation, delivering impressive improvements in both detection accuracy and efficiency.

4) Visualization of Dynamic Gate Maps: Fig. 8 visualizes and compares the dynamic gate maps in the dynamic subspace from the horizontal and vertical directions. It is observed that objects of different categories and scales are precisely highlighted while the background interference is significantly suppressed. Specifically, as the depth $\mathcal{L}$ increases, as shown in Fig. 8(a), more locations are progressively disabled to boost detection efficiency, with only the foreground determinative locations retained. Meanwhile, as the representation scales vary from P_3 to P_7 within the detection head in Fig. 8(b), the instances of various sizes are still accurately captured with fine granularity, ensuring precise and efficient detection in RS scenarios. These visualization results highlight the effectiveness of dynamic gates in establishing cross-scale head representations during dynamic learning.

5) Effect of Dynamic Scale Transformations: Table IV verifies the effectiveness of the proposed dynamic paths for scale transformations under different configurations. It can be observed that with either the forward or upward/downward paths, dynamic learning achieves improved mAPs and reduced computational costs compared to the baseline. In particular, the dynamic upward/downward paths enable scale upsampling and downsampling, thereby significantly reducing computational complexity. Utilizing both forward and upward/downward paths together yields the best results across all metrics, in terms of both accuracy and efficiency, demonstrating that these two types of dynamic scale transformations are complementary and adaptively reinforcing within the proposed dynamic learning.

6) Effect of Computational Efficiency Expectation: The impact of computational efficiency expectation on dynamic mutual optimization is directly controlled by the hyperparameter γ [see (4) and (11)] during training. The performance of our model under varying γ values is illustrated in Table V and in Fig. 9. Across all γ values, dynamic learning consistently outperforms the baseline, with $\gamma = 0.1$ yielding the highest detection accuracy across all mAP metrics. It is worth noting that the primary role of γ is to regulate the expected computational load. As γ increases, the weight of dynamic loss in dynamic mutual optimization rises, leading to a remarkable reduction in overall computational cost, albeit with a slight decrease in mAPs. These results demonstrate the effectiveness of this expectation strategy in achieving an excellent tradeoff between accuracy and computational complexity.

7) Componentwise Analysis in Mutual Learning: Table VI tabulates the performance of the two key components of the proposed mutual learning scheme: mutual prediction and mutual supervision. Notably, mutual prediction involves structural modifications by introducing mutual predictors, while mutual supervision primarily contributes to the training process by computing the mutual metric to facilitate adaptive sample assignment. The results in Table VI highlight significant improvements and the effectiveness of both components. Specifically, introducing mutual supervision alone yields accuracy gains of 1.6% mAP_{50}. Furthermore, incorporating mutual prediction significantly enhances performance, as evidenced by the complete mutual learning setup achieving an mAP_{50} of 68.6%. The integration of mutual prediction and mutual supervision within the dynamic learning context can amplify these improvements, with substantial reductions in computations, indicating their interdependence and mutual reinforcement. Overall, these observed benefits can be attributed to the

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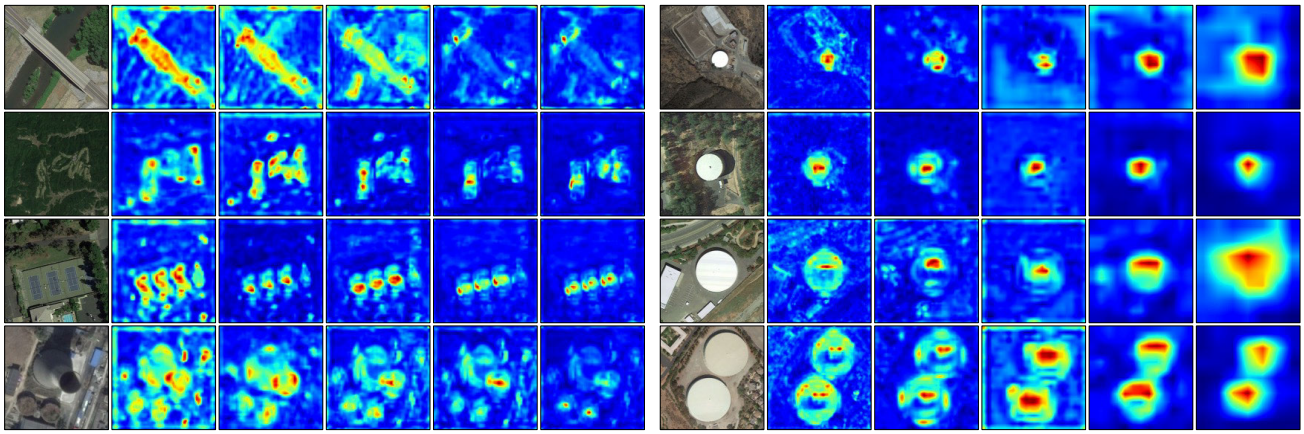


Fig. 8. Visualization of dynamic gates in the dynamic subspace. (Left to right) Gate maps in (a) and (b) correspond to the (horizontal) increase in dynamic routing depth $\mathcal{L}$ and the (vertical) increase in representation scale levels from P_3 to P_5, respectively. In (a), as the routing depth increases, the focused regions become more compact, promoting higher efficiency. In (b), several instances of the same category but with different scales are selected and compared to illustrate the high adaptability of the proposed dynamic learning scheme to the challenges posed by large-scale variations in the RS context. This also highlights its effective and comprehensive activation of representations across all scales.

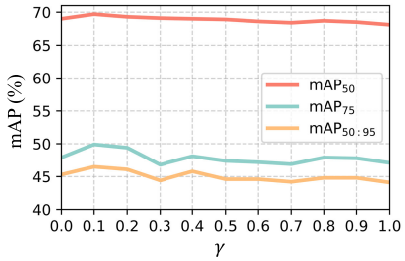


Fig. 9. Performance trend as γ is set to different values ranging from 0 to 1.0, across three mAP metrics.

TABLE VII

ANALYSIS OF DIFFERENT SETTINGS FOR THE TWO HYPERPARAMETERS α AND β IN CONSTRUCTING MUTUAL SUPERVISION

Methods	μ	ν	mAP_{50}	mAP_{75}	$mAP_{50:95}$	FLOPs
Baseline	–	–	64.1	43.4	40.9	189.6G
Mutual Learning	0.5	1	66.9	47.4	44.0	**130.1G**
	0.5	2	67.6	47.0	44.0	
	0.5	4	67.5	47.6	44.4	
	0.5	6	68.1	48.6	45.0	
	1.0	1	67.0	46.7	43.8	
	1.0	4	67.1	47.4	44.2	
	1.0	6	**68.6**	**49.2**	**45.6**	
	1.0	8	67.1	47.0	44.0	
	1.5	6	68.4	47.5	44.2	

improved consistency of subtask predictions within the mutual learning framework, further demonstrating its effectiveness.

8) Effect of Hyperparameters in Mutual Metrics: Table VII analyzes the impact of the two hyperparameters μ and ν in (8), which jointly determine the mutual metric/supervision. By varying μ and ν, minor performance fluctuations are observed, with differences of up to 1.7% mAP_{50} and 1/8% $mAP_{50:95}$. Notably, all combinations outperform the baseline, validating the effectiveness of constructing the subtask mutual supervision. Specifically, with $\mu = 0.5$, increasing ν from 1 to 6 gradually boosts mAP_{50} from 66.9% to 68.1%. The optimal detection accuracy is achieved at $\nu = 1.0$ and $\beta = 6$, which are adopted as the default settings for the experiments. Overall, the detection performance presents a low sensitivity to these two hyperparameters, demonstrating the high stability of the gains achieved through mutual supervision.

9) Effect of Different Backbones and Detectors: Tables VIII and IX demonstrate the extensive applicability of dynamic mutual learning across a variety of backbones and detectors, respectively, for RS object detection. In Table VIII, both representative CNNs and Transformers (e.g., PVT [83]) are employed as backbones, consistently validating the effectiveness of the proposed methods. For example, with ResNet101, enabling dynamic learning alone boosts mAP_{50} from 65.8% to 70.5%, and further gains to 73.1% are achieved with full dynamic mutual learning. PVT-based frameworks exhibit more remarkable efficiency advantages, offering comparable or superior accuracy to ResNet-based frameworks and benefiting further from the proposed learning paradigms. In addition, Table IX evaluates our dynamic mutual learning scheme across other prominent detection architectures beyond the baseline in RS scenarios. For example, the anchor-based detector RetinaNet [44] improves mAP_{50} from 60.9% to 67.2% while reducing FLOPs from 133.1G to 85.1G, showcasing enhanced accuracy and efficiency. Similarly, the anchor-free detectors FCOS [47] and FreeAnchor [55] exhibit mAP_{50} gains of 6.9% and 7.7%, respectively, driven by dynamic mutual learning. The results in these two tables collectively underscore the robustness and adaptability of dynamic mutual learning across diverse backbones and detectors, consistently delivering superior performance while reducing computational overhead.

10) Qualitative Comparison: Fig. 10 provides a qualitative comparison of detection results between the baseline and our proposed dynamic mutual learning method. It can be clearly observed that dynamic mutual learning consistently yields more accurate and complete detections across different object scales and categories, closely aligning with the corresponding ground truths. For instance, in the first column, while both methods provide correct classification predictions,

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TABLE VIII

PERFORMANCE COMPARISON OF DIFFERENT DETECTION BACKBONES WITHIN THE PROPOSED DYNAMIC MUTUAL LEARNING FRAMEWORK

Different Backbones	Dyna. Learn.	Mutu. Learn.	mAP_{50}	mAP_{75}	$mAP_{50:95}$	Params (M)	FLOPs (G)
ResNet101 [80]	×	×	65.8	43.2	41.0	55.6	237.1
	✓	×	70.5 (+4.7)	51.2 (+8.0)	47.7 (+6.7)	46.8	118.4
	✓	✓	73.1 (+7.3)	55.5 (+12.3)	51.6 (+10.6)	47.8	124.7
PVT-Tiny [83]	×	×	65.7	41.3	39.3	26.0	155.2
	✓	×	69.7 (+4.0)	49.8 (+8.5)	46.0 (+6.7)	**13.1**	**46.0**
	✓	✓	73.8 **(+8.1)**	54.9 (+13.6)	50.7 **(+11.4)**	14.6	52.7
PVT-Small [83]	×	×	67.7	43.2	41.3	37.3	174.2
	✓	×	71.9 (+4.2)	52.1 (+8.9)	48.3 (+7.0)	21.5	67.8
	✓	✓	**75.2** (+7.5)	**57.1 (+13.9)**	**52.7 (+11.4)**	23.1	68.1

TABLE IX

PERFORMANCE COMPARISON OF THE PROPOSED DYNAMIC MUTUAL LEARNING WHEN INSTANTIATED ON VARIOUS DETECTORS IN RS SCENARIOS

Different Detectors	Dyna. Learn.	Mutu. Learn.	Supervision Strategy	mAP_{50}	mAP_{75}	$mAP_{50:95}$	Params (M)	FLOPs (G)
RetinaNet [44]	×	×	Fixed	60.9	38.0	36.4	36.5	133.1
	✓	×	Fixed	65.6 (+4.7)	42.4 (+4.4)	40.2 (+3.8)	35.5	**70.1**
	✓	✓	Adaptive	67.2 (+6.3)	48.6 (+10.6)	45.2 (+8.8)	32.9	85.1
FCOS [47]	×	×	Fixed	60.0	35.2	34.5	31.9	123.6
	✓	×	Fixed	64.4 (+4.4)	40.9 (+5.7)	39.7 (+5.2)	**27.8**	78.3
	✓	✓	Adaptive	66.9 (+6.9)	48.5 (+13.3)	45.0 (+10.5)	28.8	75.6
FreeAnchor [55]	×	×	Adaptive	66.1	42.0	40.8	36.5	133.1
	✓	×	Adaptive	72.4 (+6.3)	49.1 (+7.1)	46.3 (+5.5)	35.5	73.5
	✓	✓	Adaptive	**73.8 (+7.7)**	**56.1 (+14.1)**	**51.7 (+10.9)**	32.9	77.4

TABLE X

DETECTION ACCURACY COMPARISON OF VARIOUS RS OBJECT DETECTION METHODS ON THE NWPU VHR-10 DATASET. THE TOP TWO BEST RESULTS IN EACH COLUMN ARE BOLDED

Methods	Backbones	AP	SH	ST	BD	TC	BC	GTF	HA	BR	VE	mAP_{50}
RICNN [6]	AlexNet	88.4	77.3	85.3	88.1	40.8	58.5	86.7	68.6	61.5	71.1	72.6
CAD-Net [84]	ResNet101	97.0	77.9	95.6	93.6	87.6	87.1	99.6	**100**	86.2	89.9	91.5
MEDNet [37]	ResNet101	99.2	94.4	82.2	98.5	95.4	95.2	98.3	88.1	75.1	89.3	91.6
GLNet [85]	ResNet101	**100**	88.7	84.4	98.5	81.6	88.2	**100**	97.2	88.4	**90.9**	91.8
CANet [86]	ResNet101	99.9	85.9	**99.3**	97.3	97.8	84.8	98.4	90.4	89.2	90.3	93.3
LGDA [87]	ResNet50	98.8	**98.4**	94.6	89.8	94.2	93.1	91.3	95.7	96.9	88.0	94.1
CoF-Net [12]	ResNet50	**100**	90.9	96.1	98.8	91.1	95.8	**100**	91.4	**89.7**	90.8	94.5
GLSA-Net [28]	ResNet50	99.9	95.8	**97.1**	**99.4**	98.8	86.1	99.5	**97.5**	84.8	86.8	94.6
AGMF-Net [88]	AGMF-Net	99.9	94.2	95.5	98.4	**99.7**	**96.3**	99.3	96.9	84.6	87.3	**95.2**
Dynamic Mutual Learning (Ours)	ResNet50	**100**	**98.7**	89.6	**99.8**	**99.5**	**99.6**	99.7	95.4	**89.9**	**96.4**	**96.9**

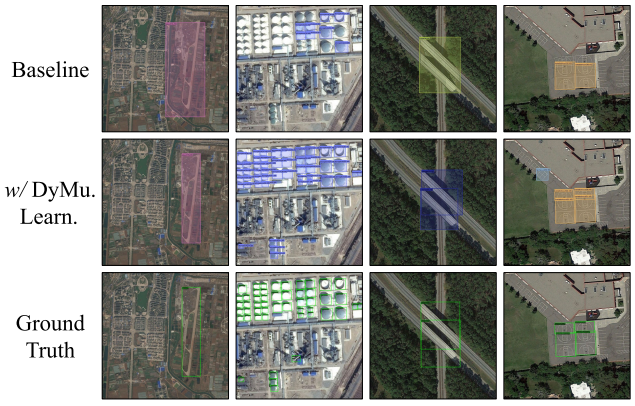


Fig. 10. Qualitative comparison of several detection results generated by the baseline and the enhanced version incorporating our proposed dynamic mutual learning on the DIOR dataset.

our method delivers more precise localization results. In the third column, the baseline method incorrectly identifies a large rectangular area as a single object with the wrong category, whereas our method successfully segments it into two adjacent instances, achieving ideal detections. Furthermore, Fig. 11 provides a comparison of detection results and corresponding feature visualizations between our method and two representative detectors, that is, RetinaNet [44] and FCOS [47], which illustrates the performance gain and its cause. In the detection results, our method produces more precise bounding boxes with fewer missed and false detections. For example, in Fig. 11(c), the competitive detectors yield multiple ambiguous detections, whereas our method delivers a clean and high-confidence detection. More fundamentally, the visualized features in Fig. 11(b) and (d) reveal the internal reason for this superiority. The activations of our method are precisely concentrated on the most discriminative object parts, while those of other methods are more diffuse and noisy. This demonstrates that our method learns more discriminative and spatially coherent representations by the proposed cross-scale dynamic interaction and classification-localization inconsistency, directly leading to more accurate and reliable

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TABLE XI

DETECTION ACCURACY COMPARISON OF VARIOUS RS OBJECT DETECTION METHODS ON THE DIOR DATASET. THE TOP TWO BEST RESULTS IN EACH COLUMN ARE BOLDED

Methods	Backbones	APL	APO	BF	BC	BR	CH	DAM	ESA	ETS	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP_{50}
CSFF [31]	ResNet101	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.0
O^2-DNet [89]	Hourglass104	61.2	80.1	73.7	81.4	45.2	75.8	64.8	81.2	**76.5**	79.5	79.7	47.2	59.3	72.6	70.5	53.7	82.6	55.9	49.1	77.8	68.4
SCRDet++ [13]	ResNet50	64.3	79.0	73.2	85.7	45.8	76.0	**68.4**	79.3	68.9	77.7	77.9	56.7	62.2	70.4	67.7	60.4	80.9	63.7	44.4	84.6	69.4
MSFC-Net [34]	ResNeSt101	85.8	76.2	74.4	**90.1**	44.2	78.1	55.5	60.9	59.5	76.9	73.7	49.6	57.2	89.6	69.2	**76.5**	86.7	51.8	55.2	84.3	70.1
GLNet [85]	ResNet101	62.9	83.2	72.0	81.1	50.5	79.3	67.4	**86.2**	70.9	81.8	83.0	51.8	62.6	72.0	75.3	53.7	81.3	**65.5**	43.4	**89.2**	70.7
AGMF-Net [88]	AGMF-Net	**90.9**	72.8	79.3	89.7	44.7	**81.4**	59.3	66.0	62.7	73.8	79.2	**65.0**	61.7	**91.7**	78.6	75.8	**90.7**	60.0	58.0	83.1	73.2
LGDA [87]	ResNet50	79.6	**83.4**	**81.8**	89.2	**52.8**	73.7	68.1	**84.5**	61.2	77.3	81.9	60.1	55.2	**89.7**	64.9	**82.7**	85.2	63.9	**58.5**	88.6	74.1
DGRN [90]	Cascade RCNN	68.1	79.6	75.1	88.1	**55.7**	**87.4**	65.8	83.7	**74.0**	**82.2**	**92.6**	**63.2**	**68.0**	76.3	**82.6**	68.2	88.9	**65.1**	**60.2**	86.1	**75.4**
Dynamic Mutual Learning (Ours)	ResNet50	**87.9**	**87.3**	**86.1**	**90.4**	49.0	**81.4**	**71.9**	83.9	71.1	**84.1**	**83.4**	51.7	**63.5**	76.4	**81.6**	74.5	**90.4**	64.5	53.1	**90.2**	**76.1**

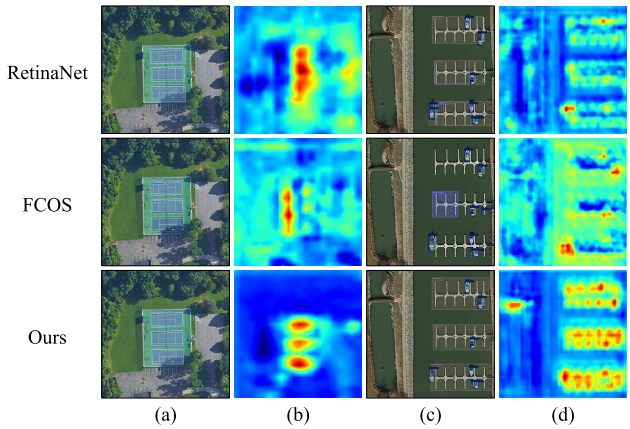


Fig. 11. Visualization comparison of detection results and features generated by different detectors on the DIOR dataset. (a) and (c) Detection results of different algorithms. (b) and (d) Corresponding features.

detections. Overall, these qualitative results in Figs. 10 and 11 highlight the significant performance gains enabled by the proposed dynamic mutual learning, validating its effectiveness in enhancing cross-scale representation capabilities and subtask consistency for RS object detection.

D. Comparison With State-of-the-Art RS Object Detection Methods

To evaluate the effectiveness and superiority of the proposed dynamic mutual learning in various RS scenarios, comprehensive comparisons are conducted against recent state-of-the-art RS detectors on two representative datasets, NWPU VHR-10 [6] and DIOR [17], respectively. As shown in Tables X and XI, our method consistently outperforms existing competitors on both datasets. Specifically, on the NWPU VHR-10 dataset in Table X, our method, employing the lightweight ResNet50 as its backbone, achieves an impressive mAP_{50} of 96.9%, outperforming numerous recent counterparts that use the same or heavier backbones. Similarly, Table XI, which tabulates the evaluation results of different methods on the more challenging DIOR dataset, demonstrates the strengths of our method, achieving an mAP_{50} of 76.1%. It surpasses other leading methods like AGMF-Net [88] and DGRN [90], which rely on more sophisticated backbones. Furthermore, the significant improvements extend beyond overall accuracy and include individual category performance. Our method secures top positions in 7 and 12 out of 10 and 20 categories on the NWPU VHR-10 and DIOR datasets, respectively, highlighting its ability to generalize effectively and adapt to varying RS contexts.

V. CONCLUSION

This article proposes a novel dynamic mutual learning paradigm for object detection in aerial imagery, aimed at improving both detection accuracy and efficiency across diverse contexts. To address limitations in existing methods—namely, subtask inconsistency at the prediction and supervision levels, as well as cross-scale impracticability at the representation level—this framework integrates dynamic learning and mutual learning schemes into a unified detection head. This integration enhances fine-grained subtask representation interaction while significantly reducing computational complexity. The promising efficiency gains demonstrated by dynamic mutual learning make it a compelling choice for practical applications in aerial scenarios. Extensive experiments conducted on the DIOR and NWPU VHR-10 datasets validate the effectiveness and superiority of the framework. The proposed detection paradigm achieves competitive accuracy and efficiency, consistently outperforming recent state-of-the-art aerial detectors. These findings highlight the potential of dynamic mutual learning to advance various tasks within the aerial context.

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